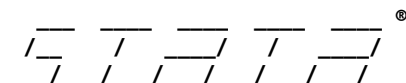


User: Vahid Karimi Motahhar
Project: TrainingForecasting



Statistics and Data Science

18.0
SE-Standard Edition

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. Maximum number of variables is set to 5,000 but can be increased;
see [help set maxvar](#).

1 . *Import and analyses time:

2 .

3 . timer on 8

4 .

5 . import excel "C:\Users\vahid\OneDrive - University of Iowa\Thesis Final Version\Submission to IJF\GJP_Training\
> t\New folder\Conditionally accepted\ReCod.xlsx", sheet("ReCod") firstrow
(39 vars, 50,161 obs)

6 .

7 .

8 .

9 .

10 .

11 . *How does training improve individual forecasts? Modeling Differences in Compensatory and Non-Compensatory Bias
> l Forecasts

12 .

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38 .
39 .
40 .
41 .
42 .
43 . *Total Timer for analyses on:

44 .
45 . timer on 9

46 .
47 . *Model 1 with timer:

48 .
49 . timer on 1

50 .
51 . set seed 6092024

52 .
53 . bayes: logit BinaryOutcome c.ForecastLogRatio if Experience == 0

```

Burn-in ...
Simulation ...

Model summary

Likelihood:

BinaryOutcome ~ logit(xb_BinaryOutcome)

Prior:

{BinaryOutc~e:ForecastLogRatio _cons} ~ normal(0,10000) (1)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian logistic regression
Random-walk Metropolis-Hastings sampling

```
MCMC iterations = 12,500
Burn-in         = 2,500
MCMC sample size = 10,000
Number of obs   = 39,481
Acceptance rate = .1657
Efficiency: min = .08137
              avg = .08514
              max = .08891
```

Log marginal-likelihood = -19772.418

BinaryOutcome	Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
ForecastLogRatio	.3428654	.0069432	.000243	.3428856	.3297828	.3571815
_cons	-1.040546	.0128871	.000432	-1.039804	-1.066064	-1.016002

Note: Default priors are used for model parameters.
Note: Adaptation tolerance is not met.

```

54 .
55 . timer off 1

56 .
57 . timer list 1
      1:      49.87 /      1 =      49.8710

58 .
59 .
60 .
61 . *Model 2 with timer:

62 .
63 . timer on 2

64 .
65 . set seed 6092024

66 .
67 . bayes: melogit BinaryOutcome c.ForecastLogRatio || user id: c.ForecastLogRatio

```

```
Burn-in 2500 aaaaaaaaaa1000aaaaaaaaa2000aaaaaa done
Simulation 10000 .....1000.....2000.....3000.....4000.....5000.....6000.....7000.....
> 000.....10000 done
```

Multilevel structure

```
user_id
{U0}: random intercepts
{U1}: random coefficients for ForecastLogRatio
```

Model summary

```
Likelihood:
  BinaryOutcome ~ melogit(xb BinaryOutcome)
```

```
Priors:
  {BinaryOutc~e:ForecastLogRatio _cons} ~ normal(0,10000) (1)
  {U0} ~ normal(0,{U0:sigma2}) (1)
  {U1} ~ normal(0,{U1:sigma2}) (1)
```

```
Hyperpriors:
{U0:sigma2} ~ igamma(.01,.01)
{U1:sigma2} ~ igamma(.01,.01)
```

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression	MCMC iterations =	12,500
Random-walk Metropolis-Hastings sampling	Burn-in =	2,500
	MCMC sample size =	10,000
Group variable: user_id	Number of groups=	851
	Obs per group:	
	min =	1
	avg =	58.9
	max =	181
Family: Bernoulli	Number of obs =	50,161
Link: logit	Acceptance rate =	.3158
	Efficiency: min =	.001603
	avg =	.01709
	max =	.0527

Log marginal-likelihood

	Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome						
ForecastLogRatio	.5603127	.0143024	.002132	.5610709	.5301141	.5878185
_cons	-.9920449	.0117032	.00051	-.9918111	-1.017125	-.9699269
user_id						
U0:sigma2	.0035771	.0015821	.000395	.0032093	.0014115	.0069379
U1:sigma2	.0833537	.0076594	.000784	.0831036	.0688809	.0997087

Note: Default priors are used for model parameters.
Note: There is a high autocorrelation after 500 lags.
Note: Adaptation tolerance is not met in at least one of the blocks.

```
68 .
69 . timer off 2

70 .
71 . timer list 2
    2:    817.62 /      1 =    817.6210

72 .
73 .
74 .
75 .
76 .
77 . *Model 3 with timer:

78 .
```

```

79 . timer on 3

80 .
81 . set seed 6092024

82 .
83 . bayes: melogit BinaryOutcome c.ForecastLogRatio##i.TrainingBinary || user_id: c.ForecastLogRatio

```

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done
Simulation 100001000.....2000.....3000.....4000.....5000.....6000.....7000.....
> 000.....10000 done

Multilevel structure

user_id	
{U0}	random intercepts
{U1}	random coefficients for ForecastLogRatio

Model summary

Likelihood:
BinaryOutcome ~ melogit(xb_BinaryOutcome)

Priors:

{BinaryOutc~e:ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:_cons}	~ normal(0,10000)	(1)
{U0}	~ normal(0,{U0:sigma2})	(1)
{U1}	~ normal(0,{U1:sigma2})	(1)

Hyperpriors:

{U0:sigma2}	~ igamma(.01,.01)
{U1:sigma2}	~ igamma(.01,.01)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression	MCMC iterations =	12,500
Random-walk Metropolis-Hastings sampling	Burn-in =	2,500
	MCMC sample size =	10,000
Group variable: user_id	Number of groups=	851
	Obs per group:	
	min =	1
	avg =	58.9
	max =	181
Family: Bernoulli	Number of obs =	50,161
Link: logit	Acceptance rate =	.2878
	Efficiency: min =	.001755
	avg =	.01554
	max =	.04418

Log marginal-likelihood

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome	ForecastLogRatio	.4900006	.0207033	.003214	.4884102	.4496442	.5338705
	1.TrainingBinary	.0159932	.0238054	.001133	.0158546	-.0312655	.0605214
TrainingBinary#c.ForecastLogRatio	1	.145275	.0283638	.004093	.1452252	.0874597	.2027102

	_cons	-.9991322	.0155361	.000995	-.9991846	-1.029551	-.9687961
user_id							
	U0:sigma2	.003008	.0011281	.000269	.00278	.0014401	.0058978
	U1:sigma2	.0785223	.0074615	.000631	.0777832	.0655406	.095103

Note: Default priors are used for model parameters.

Note: There is a high autocorrelation after 500 lags.

```

84 .
85 . timer off 3

86 .
87 . timer list 3
    3: 1255.13 /      1 = 1255.1290

88 .
89 .
90 .
91 .
92 .
93 . *Model 4 with timer:

94 .
95 . timer on 4

96 .
97 . set seed 6092024

98 .
99 . bayes: melogit BinaryOutcome c.ForecastLogRatio##i.TrainingBinary c.ForecastLogRatio##i.year2binary || user_id:
> 0

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done
Simulation 10000 .....1000.....2000.....3000.....4000.....5000.....6000.....7000.....
> 000.....10000 done

```

Multilevel structure

```

user_id
  {U0}: random intercepts
  {U1}: random coefficients for ForecastLogRatio

```

Model summary

Likelihood:

```
BinaryOutcome ~ melogit(xb_BinaryOutcome)
```

Priors:

```

{BinaryOutc~e:ForecastLogRatio} ~ normal(0,10000) (1)
{BinaryOutc~e:1.TrainingBinary} ~ normal(0,10000) (1)
{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio} ~ normal(0,10000) (1)
  {BinaryOutc~e:1.year2binary} ~ normal(0,10000) (1)
{BinaryOutc~e:1.year2binary#c.ForecastLogRatio} ~ normal(0,10000) (1)
  {BinaryOutc~e:_cons} ~ normal(0,10000) (1)
    {U0} ~ normal(0,{U0:sigma2}) (1)
    {U1} ~ normal(0,{U1:sigma2}) (1)

```

Hyperpriors:

{U0:sigma2} ~ igamma(.01,.01)

{U1:sigma2} ~ igamma(.01,.01)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression
Random-walk Metropolis-Hastings samplingGroup variable: **user_id**

MCMC iterations	=	12,500
Burn-in	=	2,500
MCMC sample size	=	10,000
Number of groups	=	851
Obs per group:		
min	=	1
avg	=	58.9
max	=	181
Number of obs	=	50,161
Acceptance rate	=	.3059
Efficiency: min	=	.002575
avg	=	.01002
max	=	.02152

Family: **Bernoulli**Link: **logit**Log marginal-likelihood

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome	ForecastLogRatio	.5361353	.0193571	.003352	.5374144	.4933332	.5723646
	1.TrainingBinary	.0127702	.0232881	.00337	.0129111	-.032876	.0560048
TrainingBinary#c.ForecastLogRatio	1	.1509684	.0266868	.004813	.1512565	.1000039	.2012054
	1.year2binary	.1326383	.0233069	.001589	.1329748	.0869525	.1785681
year2binary#c.ForecastLogRatio	1	-.0710617	.0154281	.001232	-.0712995	-.1012413	-.040983
	_cons	-1.077495	.0207381	.001682	-1.077668	-1.117566	-1.035148
user_id	U0:sigma2	.0033198	.001035	.000204	.0032881	.001584	.0057165
	U1:sigma2	.0797102	.0072334	.000612	.0793784	.0667188	.095034

Note: Default priors are used for model parameters.Note: There is a high autocorrelation after 500 lags.

100 .

101 . timer off 4

102 .

103 . timer list 4

4: 1690.34 / 1 = 1690.3350

```
104 .
105 .
106 .
107 .
108 .
109 . *Model 5 with timer:

110 .
111 . timer on 5

112 .
113 . set seed 6092024

114 .
115 . bayes: melogit BinaryOutcome c.ForecastLogRatio##i.TrainingBinary##i.year2binary if Experience == 0 || user_id:
> 0
```

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaaa done
Simulation 100001000.....2000.....3000.....4000.....5000.....6000.....7000.....
> 000.....10000 done

Multilevel structure

user_id
 {U0}: random intercepts
 {U1}: random coefficients for ForecastLogRatio

Model summary

Likelihood:
 BinaryOutcome ~ melogit(xb_BinaryOutcome)

Priors:

{BinaryOutc~e:ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.year2binary}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.year2binary#c.ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary#1.year2binary}	~ normal(0,10000)	(1)
{BinaryOutc~e:1.TrainingBinary#1.year2binary#c.ForecastLogRatio}	~ normal(0,10000)	(1)
{BinaryOutc~e:_cons}	~ normal(0,10000)	(1)
{U0}	~ normal(0,{U0:sigma2})	(1)
{U1}	~ normal(0,{U1:sigma2})	(1)

Hyperpriors:
 {U0:sigma2} ~ igamma(.01,.01)
 {U1:sigma2} ~ igamma(.01,.01)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression	MCMC iterations =	12,500
Random-walk Metropolis-Hastings sampling	Burn-in =	2,500
	MCMC sample size =	10,000
Group variable: user_id	Number of groups=	851
	Obs per group:	
	min =	1
	avg =	46.4
	max =	111
Family: Bernoulli	Number of obs =	39,481
Link: logit	Acceptance rate =	.2977
	Efficiency: min =	.001895
	avg =	.007122

Log marginal-likelihood

max = .01808

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome							
	ForecastLogRatio	.5567402	.0257831	.004536	.5553023	.5071155	.613426
	1.TrainingBinary	-.0508664	.0261608	.00443	-.0504871	-.100387	-.0009611
	TrainingBinary#c.ForecastLogRatio						
	1	.1429439	.0351722	.005885	.1436615	.0667707	.2129311
	1.year2binary	.0742375	.0337885	.002801	.0731304	.0127713	.142892
	year2binary#c.ForecastLogRatio						
	1	-.0839671	.0298656	.004676	-.0851875	-.1350712	-.0257093
	TrainingBinary#year2binary						
	1 1	.1358649	.0465298	.004422	.1348107	.0441649	.2314608
	TrainingBinary#year2binary#c.ForecastLogRatio						
	1 1	.0142111	.0216697	.004978	.0139568	-.0264743	.0564755
	_cons	-1.051382	.0224028	.001666	-1.051136	-1.094269	-1.008493
user_id							
	U0:sigma2	.003013	.0009442	.000204	.0028651	.0015461	.0054884
	U1:sigma2	.089927	.008486	.000889	.0894229	.074895	.1084871

Note: Default priors are used for model parameters.Note: There is a high autocorrelation after 500 lags.Note: Adaptation tolerance is not met in at least one of the blocks.

```

116 .
117 . timer off 5

118 .
119 . timer list 5
      5:  1783.98 /      1 =  1783.9760

120 .
121 .
122 .
123 .
124 .
125 .
126 .
127 . *Total Timer for analyses off:

128 .

```

129 . timer off 9

130 .

131 . timer list 9

9: 5597.89 / 1 = 5597.8920

132 .

133 . *Import and analyses time:

134 .

135 . timer off 8

136 .

137 . timer list 8

8: 5610.98 / 1 = 5610.9800

138 .